

E-Commerce Sales & Customer Analysis

End-to-end Data Analytics Project Report

Python • Pandas • SQL/DuckDB • Power BI • DAX • GitHub

1. Executive Summary

This project analyzes a synthetic e-commerce dataset covering 2023–2025. The workflow moves from raw transactional data through data validation, cleaning, exploratory analysis, SQL-based business analysis, Power BI data modeling, DAX measures, and an interactive four-page dashboard. The objective is to demonstrate a reproducible, business-oriented analytics workflow rather than simply create visualizations.

2. Project Scope

The analysis covers revenue and profit performance, time trends, product and category performance, customer segments and customer value, geographic distribution, discount-profitability relationships, shipping performance, and data-quality validation.

3. Dataset

Table	Rows	Purpose
customers.csv	5,000	Customer master data and segments
products.csv	504	Products, categories, costs and prices
locations.csv	150	City, state and region information
orders.csv	25,000 after cleaning	Order-level transactions
order_items.csv	43,074	Line-level quantity, revenue, profit and shipping cost
dates.csv	1,096	Calendar/date dimension for 2023–2025

4. Data Quality & Cleaning

Issue	Count	Action
Duplicate order records	3	Removed exact duplicate rows
Missing customer segments	5	Preserved; no unsupported inference
Customer-name whitespace	15	Trimmed leading/trailing whitespace
Inconsistent segment casing	10	Standardized casing
Inconsistent product-name casing	8	Standardized casing
Quantity outliers	3	Flagged; originals preserved
Revenue discrepancies	3	Flagged; originals preserved
Profit discrepancies	3	Flagged; originals preserved
Invalid foreign keys	0	No correction required
Missing shipping dates	2,506	Preserved; explained by cancelled/processing orders

Financial and quantity anomalies were intentionally preserved because the correct replacement values could not be determined reliably. This avoids silently altering source data and keeps the analysis auditable.

5. Methodology

- **Data inspection:** schemas, data types, missing values, duplicates, relationships and integrity checks.
- **Cleaning:** text standardization, exact duplicate removal, date conversion and anomaly flagging.
- **EDA:** revenue, profit, order volume, products, customers, geography, discounts and shipping.
- **SQL:** DuckDB queries directly over local CSV files.
- **Power BI:** relational model across orders, order items, customers, products, locations and dates.
- **DAX:** revenue, profit, orders, ordering customers, AOV, profit margin, category orders and product margin.
- **QA:** cross-checking key metrics and testing dashboard filters across all pages.

6. Executive Metrics

Metric	Value
Recorded Revenue	■68.49 Cr
Recorded Profit	■14.72 Cr
Orders	25,000
Ordering Customers	4,175
Average Order Value	■27,395.57
Profit Margin	21.50%

7. Key Business Findings

- Recorded revenue increased from approximately ■9.42 Cr in 2023 to ■37.57 Cr in 2025, alongside strong growth in order volume.
- Average Order Value remained relatively stable at approximately ■27K–■28K, suggesting order volume was a major contributor to revenue growth.
- Electronics generated the highest recorded category revenue, while Beauty, Toys and Fashion showed substantially higher recorded profit margins.
- Premium customers had the highest recorded revenue per customer among defined customer segments.
- The South region generated the highest recorded regional revenue, while Maharashtra recorded the highest state-level revenue.
- Recorded profit margin declined as discount levels increased; the 25% and 30% discount groups showed negative recorded profit.
- Standard shipping accounted for the largest number of orders and had the longest average order-to-shipping interval.

8. Power BI Dashboard

The final Power BI report contains four analytical pages and 28 visuals. Page 1 provides an executive overview; Page 2 focuses on product and customer performance; Page 3 covers customer and geographic analysis; and Page 4 examines discounts and shipping. Page-level slicers support interactive exploration.

Page	Focus
1 — Executive Overview	KPIs, yearly revenue, category revenue, regional revenue
2 — Product & Customer Performance	Category profit/orders, top products, monthly trend, segments
3 — Customer & Geographic Analysis	Customer value, regions, states, ordering customers
4 — Discount & Shipping Analysis	Discount-profitability and shipping performance

9. Tools & Technology

Python and Pandas were used for inspection, cleaning, validation and exploratory analysis. DuckDB was used for SQL-based business analysis over local CSV files. Power BI and DAX were used for modeling, KPI creation and interactive visualization. Excel supported initial spreadsheet-based inspection. Cursor IDE was used for development and documentation, and Git/GitHub for version control and portfolio presentation.

10. Limitations & Assumptions

- The dataset is synthetic and intended for educational and portfolio purposes.
- Recorded revenue and profit include cancelled and returned orders; they should not automatically be interpreted as realized net sales.
- Three quantity outliers and three intentional financial discrepancies were preserved as documented exceptions.
- Five customers have missing segment values and were not assigned segments without sufficient evidence.
- Shipping duration represents order date to shipping date, not final delivery time.
- Discount-profit relationships are observational and do not establish causality.

11. Project Outcome

The project demonstrates an end-to-end analytics workflow that connects technical data work with business storytelling. The final deliverable is a reproducible portfolio project containing source data, Python analysis scripts, SQL analysis, a Power BI dashboard, documentation, and GitHub version control.

GitHub repository: github.com/nksimhan-data-analyst/e-commerce-sales-customer-analysis